AIA Forecaster: Technical Report

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Abstract

This technical report describes the AIA Forecaster, a Large Language Model (LLM)-based system for judgmental forecasting using unstructured data. The AIA Forecaster approach combines three core elements: agentic search over high-quality news sources, a supervisor agent that reconciles disparate forecasts for the same event, and a set of statistical calibration techniques to counter behavioral biases in large language models. On the ForecastBench benchmark (Karger et al., 2024), the AIA Forecaster achieves performance equal to human superforecasters, surpassing prior LLM baselines. In addition to reporting on ForecastBench, we also introduce a more challenging forecasting benchmark sourced from liquid prediction markets. While the AIA Forecaster underperforms market consensus on this benchmark, an ensemble combining AIA Forecaster with market consensus outperforms consensus alone, demonstrating that our forecaster provides additive information. Our work establishes a new state of the art in AI forecasting and provides practical, transferable recommendations for future research. To the best of our knowledge, this is the first work that verifiably achieves expert-level forecasting at scale.

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1 Introduction

Forecasting is a universal problem. Any system that makes informed judgments about what may happen in the future relies on at least some degree of forecasting. In agriculture, this takes the form of predicting future crop yields and food system resilience (Bueechi et al., 2023; Tanaka et al., 2023; Paudel et al., 2021). In biotechnology and medicine, clinical trials are carefully designed to help researchers predict the impact of newly developed treatments (Qian et al., 2025; Curth et al., 2024). Elsewhere in biology, AlphaFold can be viewed as a form of forecasting protein folding structure under uncertainty (Jumper et al., 2021). In academia, grant funding is a form of forecasting which scientific discoveries will be made conditional on funding (Tohalino and Amancio, 2022). Climate and environmental science models local and long-term weather trends, which are of universal importance (Soliman, 2024; Price et al., 2025; Google DeepMind, 2024). In political science, forecasting election outcomes and the impact of policy decisions are an essential component of political calculus (Jennings et al., 2020; Wang et al., 2015; Gelman et al., 2020). Militaries make use of forecasting for threat assessment and conflict predictions (Zequeira, 2024; Probasco et al., 2025). Forecasting is the very heartbeat of financial firms, which use it to estimate future economic conditions in an uncertain world (Tsay, 2005). Wherever there is uncertainty about the future, there is forecasting.

The literature distinguishes between two high-level approaches to forecasting. The first, statistical forecasting, utilizes some combination of tabular data and simulation to develop a mathematical model of what might occur in the future. In contrast, *judgmental forecasting* is the process of aggregating unstructured data (e.g., news articles, scientific reports, etc.) and using this data in conjunction with past experience to logically predict some future outcome (Lawrence et al., 2006).

To better understand the two types of forecasting and their relationship, let us consider an example. Suppose a business wants to predict whether a certain piece of real estate will appreciate in value next year. By comparing the property to other similar properties, and looking at recent trends in the market, the company can develop a statistical model with some degree of sophistication. However, this modeling has its limits. Suppose that a news article is released, claiming that the foundation of a nearby building is shifting, causing it to slowly tilt. Even without performing additional research, a human can predict that this fact will likely put downward pressure on the property value. However, coercing this information into a form amenable to mathematical analysis is difficult. Thus, an ideal forecast would employ some abstract reasoning process which aggregates unstructured data, past experience, and logical thinking (i.e., judgment) to arrive at a sound conclusion.

This paper will focus exclusively on the judgmental forecasting problem (hereafter referred to simply as "forecasting"). Recently, there has been an uptick in research which studies the judgmental forecasting problem, due in large part to the ubiquity of Large Language Models (LLMs). By giving LLMs access to news databases and asking them to forecast future events, it is possible to systemically evaluate the forecasting abilities of machine